SDC2026_ICEMAN_Preliminary

1. Team and project overview

Field	Response
Team Name	ICEMAN
Selected Domain	Domain 1: The Aiversecity Future-Ready Workforce
Selected SDG Track (1-17)	SDG 4 Quality Education (targets 4.4, 4.3), with SDG 8 Decent Work and Economic Growth (targets 8.2, 8.5)
Project / Solution Name	Future-Ready RPG
Name (Leader)	Eashwar Singh Sidhu
Email & Phone (Leader)	eaergfyl@gmail.com · 016-931 0279
University Affiliation(s)	Asia Pacific University
Number of Team Members	5

2.1 The problem context

Malaysian workers are being asked to reskill for AI without knowing which skills will matter. 56% of professionals in Malaysia already use AI tools at work, yet only 26% have had formal training (Hays Asia Salary Guide 2025, quoted in the Domain 1 brief). The Ministry of Science, Technology and Innovation says more than 600,000 workers will need reskilling within three to five years.

We focus on one person at one moment: a working adult whose job is changing, such as a customer service executive, an accounts clerk, a production operator or an electrician, who has about six hours a week to study and must decide what to learn next. Today that person faces three gaps:

- **No signal.** Advice about "future skills" is a static list, out of date when published, with no source or date.
- **No fit.** Nothing tells them which future roles are closest to the skills they already have, or what exactly is missing.
- **No path.** Course catalogues are long and unordered. They cannot tell how long a switch takes or what it costs.

The cost is economic (displacement and lost income), social (people without career advisers fall further behind, widening inequality) and environmental: many of the fastest-growing roles are in clean energy and efficient manufacturing, and they go unfilled when workers cannot see a route into them. A wrong guess costs months of evenings, so the decision deserves real evidence

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2.2 The proposed AI solution

Future-Ready RPG turns real, dated trend data into an interactive career skill tree a worker can play through. It guides a player from "who I am today" to ranked future roles and a quest-like learning pathway in minutes, with every stat citing its source and date.

1. **Adventurer Creation (Profile Intake):** Players pick a starting class (a sample worker), select skills, or upload a CV (.txt/Word). Keyword rules pre-tick skills with listed evidence, while an AI model suggests bonus skills for player confirmation.

2. **Future Skills Radar:** All 39 skills feature momentum stats calculated from job postings (Arbeitnow, Remotive) and arXiv research data via an ingestion pipeline. Sparse data points are dropped and flagged.

3. **Class Progression & Role Matching:** 16 future roles (mapped to O*NET occupations) are ranked by an importance-weighted match score, highlighting the main boss-level skill gap to overcome.

4. **Reskilling Quests (Pathway):** Missing skills are structured into prerequisite skill trees paired with verified free/low-cost courses, total hours, costs, and estimated completion timelines based on study pace.

5. **SDG Badges & Achievements:** Skills and roles link directly to UN SDG targets with clear rationales. Unlocking skills earns collectible SDG badges.

6. **World Scenarios:** Slow, moderate, and fast AI disruption scenarios show role outlook ranges as dynamic campaign conditions.

7. **Ask Your Guide:** An chat that explains player stats and transition pathways in English or Bahasa Malaysia.

AI powers player onboarding while keeping game logic honest: scoring is 100% deterministic and explainable. The LLM (self-hosted Qwen3-8B) only reads computed stats or redacted CVs, withholding answers referencing unverified numbers.

Workshop skills applied: GenAI APIs (RunPod Serverless endpoint), database design (SQLite snapshots), and cloud hosting (Replit Deployments).

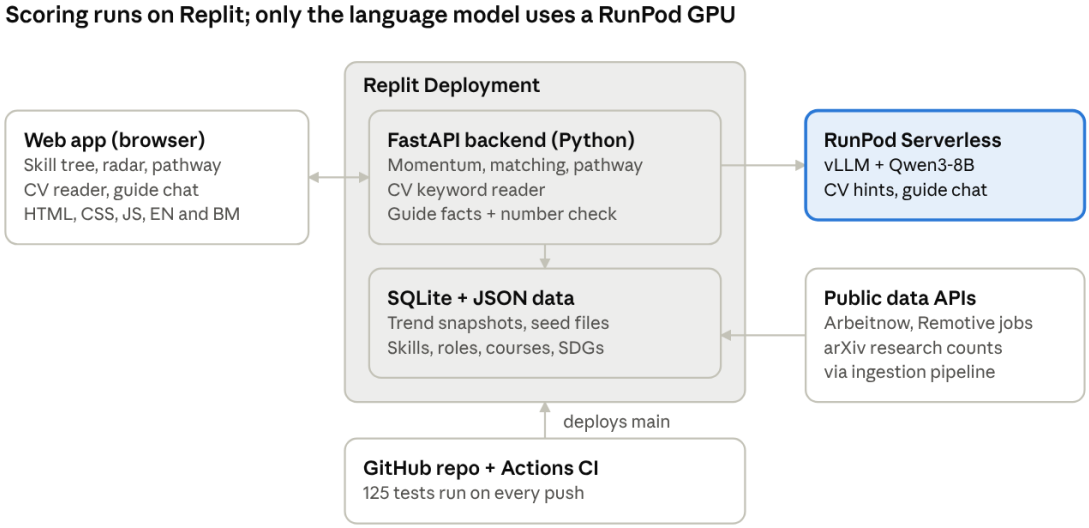
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3.1 Conceptual tech stack

The prototype is built and deployed. One FastAPI app on Replit serves the web app and all scoring, and only the language model runs on a GPU, on RunPod.

Layer	Technology	What it does
Frontend	HTML, CSS, vanilla JavaScript (ES modules), no build step	Skill tree, radar, pathway, CV reader, guide chat, EN/BM toggle

Layer	Technology	What it does
Backend	Python 3.12, FastAPI, Uvicorn, Pydantic	API, momentum scoring, role matching, pathways, CV reading, guide facts
Database	SQLite, JSON data files checked against JSON Schema	Trend snapshots; skills, roles, courses and SDG map
Data sources	Arbeitnow and Remotive job APIs, arXiv API, O*NET 31.0	Job-demand and research signals; occupation basis for roles
AI model	Qwen3-8B served with vLLM on a RunPod Serverless endpoint	CV skill suggestions and "Ask your guide" answers
Hosting	Replit Deployments	Live prototype at sdggez.replit.app
Dev and CI	GitHub, GitHub Actions, pytest	125 automated tests on every push



System architecture · Replit app, RunPod model, public data APIs

The browser calls the FastAPI backend; the backend reads trend snapshots and data files, calls the RunPod endpoint only for CV hints and guide answers, and keeps working if RunPod is off. The ingestion pipeline pulls the public APIs into new snapshots.

Sponsor-provided GenAI API or cloud platform: yes. Replit Core (provided to each participant) hosts development and the live deployment. The RunPod credit (US$25 per team) runs the Qwen3-8B endpoint. We did not use a sponsor GenAI API, the model is open source and self-hosted on RunPod.

3.2 Design-thinking and deployment strategy

- **Empathise.** We wrote four synthetic workers from the brief's user groups (customer service, accounts, production, electrical) with real job titles and study hours, and designed every screen around one question: what should I learn next?
- **Define.** One person, one moment of need, three gaps: signal, fit and path.

- **Ideate.** A game skill tree makes prerequisites, progress and SDG impact visible at a glance, instead of another job list or chatbot.
- **Prototype and test.** We built in small daily slices, each deployed and tested. Team testing changed the design: a first-visit tour because the page felt overwhelming, a grouped skill checklist, Word CV upload, and dropping trend figures based on too few postings.

Deployment readiness. The app is live on Replit Deployments, deployed from the main branch, where GitHub Actions runs 125 tests on every push. Real trend data ships with the repo, so a fresh deployment never shows mock data. Keys stay in Replit Secrets, and every AI feature falls back gracefully if RunPod is unavailable.

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4. Preliminary environmental and SDG impact framework

Our baseline is a 30-point gap between workers who use AI and workers trained for it; our KPIs measure whether a worker leaves the app with a concrete, verified next step.

Metric category	Our response
Target SDG sub-target(s)	**Target 4.4:** "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship." Supporting: **8.5** (full and productive employment and decent work for all) and **4.3** (access to technical and vocational education).
Current problem baseline	56% of Malaysian professionals use AI tools at work but only 26% have had formal training: a **30-percentage-point training gap** (Hays Asia Salary Guide 2025). More than **600,000 workers** need reskilling within three to five years (Ministry of Science, Technology and Innovation). In our prototype, verified free or low-cost courses cover **8 of 39 skills (21%)**, so most pathway steps still end without a course.
Core measurable KPI	(1) **% of tested users who reach a ranked future role and a pathway with at least one verified course within 5 minutes** of opening the app. (2) **% of skills in the catalogue with a verified free or low-cost course.**
Phase 2 impact target	(1) **80%** of at least 10 test users (working adults or final-year students) reach a role and a verified pathway within 5 minutes. (2) Course coverage from **21% to 80%** of skills (at least 32 of 39), prioritising HRD Corp-claimable programmes where the provider confirms eligibility.

Sources: Hays Asia Salary Guide 2025 and the Ministry of Science, Technology and Innovation figure, both as quoted in the SDC Domain 1 brief; UN SDG Goal 4 targets and Goal 8 targets; course coverage counted from `data/courses.json` in our repository on 24 October 2026.

5. Eligibility and readiness checklist

- ☒ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☒ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☒ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☒ All fields in this template have been completed within the word limits stated.

6. IP and originality declaration

6.1 Pre-existing / prior-owned work

No. The repository's first commit is on 20 October 2026, after the hacking period opened on 19 October 2026

6.2 Third-party code, datasets, or models

Yes. All are open source or public, declared here and credited in the app footer or `docs/sources.md`.

Name	Type	License
FastAPI	Code (web framework)	MIT
Uvicorn	Code (web server)	BSD-3-Clause
Pydantic	Code (data validation)	MIT
httpx	Code (HTTP client)	BSD-3-Clause
feedparser	Code (arXiv feed parsing)	BSD-2-Clause
PyYAML	Code (story files)	MIT
APScheduler	Code (scheduling)	MIT
jsonschema	Code (data validation)	MIT
pytest	Code (tests)	MIT
vLLM, via RunPod worker-vllm	Code (model serving)	Apache 2.0 (vLLM); MIT (worker-vllm)
Qwen3-8B	Model (pre-trained, not fine-tuned)	Apache 2.0
O*NET 31.0 Database	Dataset (occupation reference)	CC BY 4.0, credited and adapted
Arbeitnow job board API	Dataset (job postings)	Free public API, Arbeitnow terms of service
Remotive API	Dataset (job postings)	Free API; Remotive is named and

Name	Type	License
		linked back
arXiv API	Dataset (research paper counts)	arXiv API Terms of Use
Dreamstime dice pattern (image 259770746)	Image (page background)	Royalty-free license via a Dreamstime download; credited in the footer
Class illustrations (Bard, Wizard, Paladin, Druid)	Images	Vector shapes generated as code with AI assistance (Claude, by Anthropic), then reviewed by the team

6.3 Employment-related IP conflicts

No.

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Signature	 [Eashwar Singh Sidhu,]
Date	24/10/2026